

FELIX MA

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Education

University of British Columbia - GPA: 3.7/4.0

Expected Grad: May 2027

Bachelor of Applied Science in Computer Engineering

Vancouver, BC

Relevant Courses: Data Structures and Algorithms (C++), Operating Systems (C), Digital Systems Design (SystemVerilog, Quartus, FPGA Programming), Software Engineering (Java)

Work Experience

UBC Robotics and Control Laboratory

January 2026 - September 2026

Research Software Engineer Intern

Vancouver, BC

- Designed and implemented **Python** class hierarchy to increase supported ultrasound platforms from **1** to **any amount**
- Built custom beamformer in **Python**, eliminating the need to install previous library's **148**-package conda environment
- Wrote **MATLAB** scripts using **K-Wave** to generate **10+** simulated phantoms, increasing data points by **10x**
- Built live strain viewer end-to-end with **AI (Claude/Copilot)**, validating feasibility via rapid prototyping

Transoft Solutions

May 2025 – December 2025

Software Engineer Intern

Vancouver, BC

- Implemented **C++** test generation framework with **MFC** frontend, expanding test coverage from **15%** to **100%**
- Built **Qt/Python** tool adopted by QA team, cutting time to compile a list of **50-100** test vehicles by **95%**
- Identified **5** edge cases in **GTest** suite, adding coverage that uncovered **10** vehicles with faulty parameters

Excelar

September 2024 – April 2025

Software Engineer Intern

Burnaby, BC

- Built **Angular** user management dashboard, consolidating **15** pages into **1**
- Designed multi-page deployment flow for medical AI models, enabling **10+** non-technical staff to self-serve deployments
- Resolved **90+** front-end tickets via **Angular/RxJS** fixes, cutting page load times by **50%**

Technical Projects

Custom x86 Operating System — *x86 Assembly, C, NASM, OpenWatcom, Bochs*

April 2026 – August 2026

- Wrote two-stage **x86 Assembly/C** bootloader with a custom **FAT12** parser to load the kernel into RAM
- Implemented **GDT/IDT/ISR** infrastructure for 32-bit protected mode and interrupt handling
- Built physical memory manager with **paging**, using a bitmap allocator over BIOS **e820** map for **128MB** RAM

RC4 Encryption Breaker — *SystemVerilog, FPGA, Quartus/ModelSim*

June 2024

- Wrote **SystemVerilog** code to read/write encrypted message to **FPGA** RAM
- Designed **5 SystemVerilog** finite state machines (5-12 states each) to brute-force crack encrypted messages
- Verified **FPGA** design correctness via test benches and full-module simulation in **ModelSim**

FPGA Tone Organ — *SystemVerilog, FPGA, Quartus, SignalTap*

May 2024

- Built **SystemVerilog** frequency-divider module to convert a clock signal into audio frequencies
- Synthesized **SystemVerilog** via **Quartus** and programmed **FPGA** to function as a tone player
- Validated frequency and switch behavior using **SignalTap** and waveform simulation

Student Design Team

UBC LaunchPad, Student Design Team

September 2025 – April 2026

Software Developer

Vancouver, BC

- Integrated **NextJS** calendar component, streamlining volunteer scheduling for BC Brain Wellness Program
- Implemented skeleton-loading components across **9** pages, improving perceived load speed in **NextJS** app

Technical Skills

Languages: C++, Python, Typescript, C, Java, SystemVerilog, MATLAB

Frameworks/Libraries: Matplotlib, Plotly, K-Wave

Developer Tools: Linux, Git, Visual Studio, Jira, GitHub, GitLab, Quartus